

Aditya Kulkarni

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| Raleigh, NC (Open to Relocation)

PROFESSIONAL SUMMARY

MS CS candidate (NC State, May 2026) with production experience in distributed systems, scalable backend engineering, and full-stack TypeScript/Python development. At Isomer AI, owned end-to-end delivery of 10+ backend features across a production claims platform -- 394 commits, 7 deployments, real-time event pipelines, and multi-model AI agent tooling using Temporal.io workflow orchestration. Built event-driven systems handling 2,000 async requests/second and distributed ETL pipelines at Capgemini with Kafka + Spark. Proficient in Java, Python, TypeScript, Docker, PostgreSQL, and AWS; experienced in CI/CD, fault-tolerant architecture, and production observability.

CORE COMPETENCIES

Distributed Systems

Java / Python / TypeScript

Event-Driven Architecture

Cloud-Native / Microservices

Docker / Kubernetes

PostgreSQL / MySQL / MongoDB

CI/CD & Observability

Scalable Backend Systems

WORK EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern — Chicago, IL (Remote)

- Shipped **10+ backend features across 7 production deployments** -- RBAC, real-time event pipelines (Pusher), document processing (PDF, email via Microsoft Graph API, S3), and scheduled jobs; TypeScript, Node.js, PostgreSQL, Redis, Docker, GitHub Actions CI/CD, Sentry observability -- 394 commits reviewed
- Implemented **multi-provider LLM agent tooling** across Anthropic Claude, OpenAI GPT-4o, and Google Gemini using Temporal.io workflow orchestration and MCP; provider-agnostic abstraction with retry/fallback logic for production insurance claims automation

NC State University Apr 2025 - Present

Graduate Research Assistant — Raleigh, NC

- Designed a cloud-based genomic data platform on Google Cloud Run serving 500GB datasets to 100+ researchers; applied hyperparameter tuning and time-series analysis improving ML model accuracy by 12%

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern — Mumbai, India

- Built 5+ Python AI workflows including health claim automation (OCR, NLP, ML); improved accuracy by 25%; automated deployments on AWS (EC2, S3) and GCP with Grafana monitoring dashboards

Capgemini Jun 2023 - Aug 2023

AI/ML Intern — Mumbai, India

- Built distributed **ETL pipelines** with Hadoop + Spark (batch) and Kafka (real-time streaming); reduced ticket-processing latency by 20% and overall processing time by 15%

PROJECTS

Inventory Reservation Service Java 21

Transactional inventory reservation API preventing overselling via PostgreSQL row-level locking (SELECT FOR UPDATE) across multi-SKU reservations. Idempotent POST endpoints with RFC 9457 error responses; automated expiry via scheduled cleanup job. Concurrency verified with Testcontainers -- two concurrent reservations against quantity 1 produce exactly one success and one 409 Conflict. Flyway migrations, Spring Security (RBAC), multi-stage Docker build, GitHub Actions CI.

Java 21, Spring Boot 3.5, PostgreSQL, Flyway, Spring Data JPA, Spring Security, Testcontainers, Docker

AlgoPulse – Agentic AI Orchestration Platform Event-Driven

Identified API-polling bottleneck under load; re-architected to event-driven model -- Supabase Edge Functions stream AI reasoning loops directly to clients. Temporal.io manages durable workflow state; Kafka handles worker coordination and fault-tolerant message brokering: **2,000 async inference requests/second** without state loss.

TypeScript, Node.js, React, Supabase Edge Functions, Temporal.io, Kafka

PulseRank – Real-Time Leaderboard & Activity Feed Multi-DB

Event-driven platform: MySQL for transactional writes (idempotency key, outbox pattern), MongoDB for feed projections (TTL expiry), Redis sorted sets for sub-millisecond leaderboard reads. Live updates via Server-Sent Events. Testcontainers multi-store consistency tests, k6 benchmarks, OpenTelemetry + Prometheus observability, GitHub Actions CI/CD.

TypeScript, NestJS, Next.js, MySQL, MongoDB, Redis, Docker, OpenTelemetry, Prometheus

LLM Eval Harness Python

Vendor-neutral benchmark comparing Claude Sonnet 4.6, GPT-4o, and Gemini 2.5 Pro across 2,000-example frozen test set. Metrics: ExactMatch, TokenF1, grounded hallucination rate, p50/p95 latency, cost/call with paired bootstrap CIs and McNemar testing.

Python, Anthropic/OpenAI/Google SDKs, DuckDB, Parquet, Streamlit, Docker

EDUCATION

North Carolina State University – MS Computer Science Aug 2024 - May 2026

Courses: Algorithms, Operating Systems, Artificial Intelligence, Data Science, Software Engineering, Object-Oriented Development

Dwarkadas J. Sanghvi College of Engineering – BTech Computer Science Aug 2020 - May 2024

CGPA: 9.32/10 · Courses: DS & Algorithms, OS, Machine Learning, Big Data Infrastructure, Software Engineering

SKILLS

Languages: Java, Python, TypeScript, JavaScript, SQL, C, R

Backend & Systems: Spring Boot, Node.js, NestJS, Temporal.io, Kafka, Spark, Hadoop, REST APIs

Databases: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB (familiar)

Cloud & DevOps: AWS (EC2, S3), GCP (Cloud Run), Docker, Kubernetes, GitHub Actions, Terraform, OpenTelemetry, Prometheus, Grafana, Sentry

AI/ML: Anthropic Claude, OpenAI API, LangChain, MCP, Hugging Face, TensorFlow, PyTorch, RAG